M.Sc. 3rd Semester

MCS-1911: Soft Computing Techniques Using Neural Networks

Time: 3 Hours

Max. Marks: 80

Note: Attempt **One** question each from Units-I, I, II and IV weightage is 16 marks each, subparts weightage is 8. Attempt all questions in Unit-V weightage is 2 marks each.

UNIT 1

Q.1 a) Explain various Neural Network Architectures.

b) Describe various activation functions used in neural networks along with their curves.

Q.2 a) Why are hybrid systems called embedded systems? Explain various kinds of Hybrid systems.

b) Discuss supervised and unsupervised learning with examples.

UNIT 2

Q.3 a) Draw the difference between supervised and unsupervised learning.

b) How does the back preparation algorithm work? Give an example.

Q.4 a) Write a step by step algorithm for Kohonen self organizing map and explain with an example.

b) Discuss ART1 architecture with help of a diagram.

UNIT 3

Q.5 a) How fuzzy sets are different from classical sets? Explain various operations on fuzzy sets.

b) Explain the architecture and operation of fuzzy logic control systems.

Q.6 a) What is defuzzification? Explain various ways to defuzzify.

b) Discuss four methods of fuzzification and four methods of defuzzification with examples.

UNIT 4

Q.7 a) Define genetic algorithm? How is it different from the traditional system? Explain its basic concepts with examples.

b) Explain various genetic operators in detail.

Q.8 a) Give biological background of genetic algorithm? Also explain genes, fitness and population in detail.

b) How is genetic programming different from traditional programming? How can genetic programming solve optimization problems? Discuss applications of genetic programming.

UNIT 5

Write short note on

a) Mutation

b) Genotype and phenotype

c) Neuron

d) Applications of neural networks

e) Defuzzification

f) Application of fuzzy logic

g) ART

h) Perceptron

M.Sc. (Hons. School) IIIrd Semester
MCS – 1912: Software Project Management
14/12/2023

Time Allowed: 3 hours **Max. Marks: 80**

Note: Attempt **FIVE** questions in all, selecting **ONE** question each from Sections A, B, C and D. Section E (Q.No. 9) is compulsory. All questions carry equal marks.

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Section A

Q1. (a) Briefly describe the three distinct phases of the development process through which a typical software project goes. List and explain various problems with software projects.
(b) What is cost-benefit analysis in projects? What is its importance? Discuss the steps involved in performing cost-benefit analysis. (8, 8)

Q2. (a) What are the different processes involved in a project? How are planning processes different from control processes?
(b) Name and explain the features of any four Open Source Software tools for Project Management. (8, 8)

Section B

Q3. (a) What do you understand by stepwise project planning? What are the main steps in this type of project planning?
(b) Discuss the similarities and differences between PERT and CPM. How does Critical Path Analysis help in expediting the projects? (8, 8)

Q4. (a) How does network diagrams help in Project Scheduling and Resource Allocation?
(b) What is meant by Scope Management in planning a project? Discuss the methods for selecting the projects. (8, 8)

Section C

Q5. (a) Differentiate between the following:

- Process and Project Metrics
- Class-oriented metrics and Operation-oriented metrics
- Function oriented metrics and Extended function point metrics (16)

Q6. (a) What are object-oriented metrics and what are their characteristics? What are the different metrics used to verify design of an object-oriented software?
(b) State the purpose of Software Requirement Specification (SRS). Give an account of metrics for SRS quality. (8, 8)

Section D

Q7. (a) What is ERP? Describe various technologies related to ERP. Enlist and explain various application areas of ERP.

(b) What is meant by What-if Analysis and Sensitivity Analysis? (9, 7)

Q8. Write short notes on the following:

(i) Life cycle of ERP
(ii) Components of DSS
(iii) Executive Information Systems
(iv) Knowledge Management Systems

(4 X 4 = 16)

Section E

Q9. **Attempt all the parts in brief:**

(a) Who are the stakeholders of a project?
(b) Explain the term "*Project Charter*".
(c) What are the different types of activities in a project?
(d) Explain "*Work Breakdown Structure*".
(e) Briefly explain the relationship among software process, software project and software product.
(f) Define Business Process Re-engineering.
(g) What is the importance of Goal seeking Analysis.
(h) Discuss the role of a Project Manager.

(8X2 = 16)

Paper Code: MCS1914 **Paper Title: COMPUTER BASED OPTIMIZATION TECHNIQUES**
M.Sc. (H.S.) Computer Science- 3rd Semester

Time : 3 Hrs Max. Marks: 80

Note: (i) The Question Paper consists of Four UNITS.
(ii) The students are required to attempt one question from each UNIT and the compulsory question.
(iii) All questions carry equal marks.
(iv) **The students can use only Non-Programmable and Non-Storage type Calculator.**

UNIT- I

1. A manufacturer can produce four products using three resources namely labour, raw material and machine time. The unit contribution and the technological constraints are given below in the form of an LPP: 16
Max $Z = 3x_1 + 3x_2 + 4x_3 + 7x_4$, subject to $x_1 + x_2 + x_3 + x_4 \leq 9$; $5x_1 + 3x_2 + 2x_3 + x_4 \leq 60$;
$x_1 + 3x_2 + 5x_3 + 8x_4 \leq 50$; $x_1, x_2, x_3, x_4 \geq 0$.
i) Find the optimal product mix.
ii) Suppose a fifth product represented by variable x_5 is to be included in the optimal product mix, what should be its unit contribution if the resource requirements for this product are 7/4 units, 3 units and 7 units of labour, material and machine-time respectively?

2. i) Use dual simplex method to solve:
Minimize $Z = x_1 + 2x_2 + 3x_3$,
subject to: $2x_1 - x_2 + x_3 \geq 4$, $x_1 + x_2 + 2x_3 \leq 8$, $x_2 - x_3 \geq 2$, $x_1, x_2, x_3 \geq 0$.
ii) Use duality to solve the problem:
Minimize $Z = x_1 - x_2$, subject to $2x_1 - x_2 \geq 2$, $-x_1 + x_2 \geq 1$, $x_1, x_2 \geq 0$.

UNIT - II

3. A product is manufactured by four factories A, B, C and D. The unit production costs are Rs. 2, Rs. 3, Re. 1 and Rs. 5 respectively. Their daily production capacities are 50, 70, 30 and 50 units respectively. These factories supply the product to four stores P, Q, R, and S. The demand made by these stores are 25, 35, 105 and 20 units respectively. Unit transportation cost in rupees from each factory to each store is given in the following table: 16

	P	Q	R	S
A	2	4	6	11
B	10	8	7	5
C	13	3	9	12
D	4	6	8	3

Determine the extent of deliveries from each of the factories to each of the stores so that the total cost is minimum.

4. i) Maximize $Z = 2x_1 + 20x_2 - 10x_3$, 10
subject to: $2x_1 + 20x_2 + 4x_3 \leq 15$, $6x_1 + 20x_2 + 4x_3 = 20$, $x_1, x_2, x_3 \geq 0$ and are Integers
ii) Four new machines M_1, M_2, M_3 and M_4 are to be installed in a machine shop. There are five vacant places-A, B, C, D and E available. Because of limited space, machine M_2 cannot be placed at C and M_1 cannot be placed at A. C_{ij}, the assignment cost of the machine i to place j in rupees is shown below:

	A	B	C	D	E
M_1	4	6	10	5	6
M_2	7	4	-	5	4
M_3	-	6	9	6	2
M_4	9	3	7	2	3

Find the optimal schedule to assign the machine to the vacant places.

UNIT - III

5. (i) Define travelling salesman Problem. Write the steps to solve the problem using any method. 8

5. (ii) Solve the following travelling salesman problem so as to minimize the cost per cycle:

City	1	2	3	4	5
1	-	10	25	25	10
2	1	-	10	15	2
3	8	9	-	20	10
4	14	10	24	-	15
5	10	8	25	27	-

6. (i) Define DPP? Describe an algorithm to solve the DPP.

(ii) A firm has divided its marketing area into three zones. The amount of sales depends upon the number of salesman in each zone. The firm has been collecting the data regarding sales and salesman in each area over a number of past years. For the next year firm has only 9 salesman and the problem is to allocate these salesmen to three different zones so that total sales are maximum.

No. Of salesmen	Zone 1	Zone 2	Zone 3
0	30	35	42
1	45	45	54
2	60	52	60
3	70	64	70
4	79	72	82
5	90	82	95
6	98	93	102
7	105	98	110
8	100	100	110
9	90	100	110

UNIT - IV

7. (i) Define queuing model. What are its applications? Define the structure of basic queuing system

(ii) A project is composed of 7 Jobs whose time estimation are given below:

Activity	1-2	1-3	1-4	2-5	3-5	4-6	5-6
T_m	7	16	7	9	20	14	2
T_o	8	18	9	10	24	16	3
T_p	9	20	11	11	28	18	4

(a) Draw the network and calculate the length and variance along the critical path.

(b) Find the probability of completing the project one day earlier and 2 days later.

8. The arrival of customer and service times of customer are having the following distributions. Simulate this queuing system for 10 periods by using the following random numbers and calculate mean waiting time and mean queue length.

Inter-Arrival Time(min)	Probability
5	0.15
6	0.35
7	0.40
8	0.10

Service Time (min)	Probability
7	0.10
8	0.35
9	0.45
10	0.10

Random Num for arrival: 36, 60, 82, 14, 14, 62, 62, 10, 55, 14.

Random Number for service: 34, 35, 31, 62, 40, 73, 88, 70, 19, 40.

UNIT – V

9. Answer the following questions briefly:

(i) Define basic variables.

(ii) What do you understand by unbalanced transportation problem?

(iii) What is PERT?

(iv) What are the advantages of simulation?

(v) Define binary linear programming.

(vi) Write the steps to write a dual of a given problem.

(vii) What do you understand by the term sensitivity analysis?

(viii) What is the advantage of using revised simplex method?

M.Sc. 1st Semester (Data Science Specalization)

MDS2404: Operating System with Linux *December 17, 2024*

Note: Attempt FIVE Questions in all, including Q-9 in SECTION-E, which is compulsory, taking ONE each from Section-A to Section-D.

Time: 3 Hours *Max Marks: 80*

SECTION-A

1)

a) Identify different services that you would expect to be provided by a typical OS. List the major features of the Distributed operating systems.

b) Draw the state diagram of a process from its creation to termination, including all transitions, and briefly elaborate every state and every transition. Explain using Linux OS. [8, 8]

2)

a) What is process synchronization? Describe the Reader-Writer problem in the context of process synchronization:

b) Compute average waiting time given the *Round Robin scheduling* algorithm with time quantum = 3 and (P1, 0, 5), (P2, 2, 3), (P3, 3, 2), (P4, 4, 5), (P5, 5, 4) where first component of the triple is the process number, the second component is the time arrival and the third component is the burst time. Use FCFS as tiebreaker if needed. [8, 8]

SECTION-B

3) Define system deadlock. List and describe four conditions necessary for deadlock to occur in an operating system. Give an example of a simple resource deadlock involving three processes and three resources. Draw the appropriate resource allocation graph. [16]

4) What is disk scheduling? How is it different from disk management? Assume that the disk head is currently at cylinder zero (0). (Where the total cylinders are from 0 to 199). The disk queue requests are 75, 120, 155, 80, 170, 60, and 100. Calculate the total distance (in cylinders) moved by the disk arm by the *SCAN*, *LOOK* and *SSTF* algorithms. [16]

SECTION-C

5)

a) How is Linux shell different from Linux kernel? Elaborate the salient features of *Bourne Again Shell (BASH)*.

b) Give an account of the following in the context *BASH*:

i. File and Directory management Features

ii. Process Management Features [8, 8]

6)

a) Write a shell script to check whether a given string is a palindrome or not (without actually reversing the string).

b) Write a shell script that reads the input file, translates all the characters in the input file into the uppercase, and then stores the results in the specified output file. Synopsis: *upper –i infile –o outfile* [8, 8]

SECTION-D

7)

a) What is Linux system administration? Discuss the role of '*tar*' facility and '*gzip*' facility for system administration.

b) How do you administer user and group accounts in Linux? List the system files used to store the user and group account information, and the initialization files associated with user accounts. [, 8]

8)

a) Write a 'C' program that illustrates the creation of child process using '*fork()*' system call. One process finds sum of even series and other process finds sum of odd series.

b) Write a 'C' program that uses *fork( ), pipe( ), dup( ), and close( )* system calls to simulate/execute the shell command as given: *more myfile.txt | sort*, where | is a piping operator. [8, 8]

SECTION-E (Compulsory Question)

9)

a) In an '*ext2*' Linux file system, an inode consists of 12 Direct Disk Block Addresses (DBA), 1 single indirect DBA, 1 doubly indirect DBA and 1 triple indirect DBA. The DBA requires 32 bits and the Disk block size is 1 KB. Then what is the maximum file size possible?

b) What are the differences between I/O bound and CPU bound processes? Describe one typical application that is I/O bound; describe one that is CPU bound.

c) What is a Linux file system? Give difference between '*ext2*' and '*ext3*' file system architectures.

d) Explain *execl()* and *execlp()* system calls with examples. [4, 4, 4, 4]

M.Sc.(Hons.) IIIrd Semester
MCS-1912: Software Project Management
Final Exam: December 12' 2024

Time Allowed: 3 hours **Max. Marks: 80**

Note: Attempt five questions in all selecting one question each from units I, II, III and IV. Q.No. 9 is compulsory. All questions carry equal marks.

UNIT-I

Q1. a) Explain the stages of a software project life cycle with examples. **(8)**
b) Discuss the challenges commonly faced in software projects and how effective project management can mitigate them. **(8)**

Q2. a) Define and explain the importance of periodic status assessments and checkpoints in software projects. **(8)**
b) Describe the process interactions involved in initiating, planning, executing, controlling, and closing a software project. **(8)**

UNIT-II

Q3. a) What is integration management? Explain the steps involved in developing and executing a project plan. **(8)**
b) Define scope statement and project charter. How do they contribute to project planning? **(8)**

Q4. a) Discuss the significance of sequencing and scheduling activities in time management for a project. **(8)**
b) Explain Program Evaluation and Review Technique (PERT). Discuss how PERT differs from critical path analysis in project scheduling. **(8)**

UNIT-III

Q5. a) Discuss extended function point metrics and their advantages over traditional size-oriented metrics. **(8)**
b) Discuss metrics for software testing and maintenance, and explain how they help in improving software quality. **(8)**

Q6. a) What are metrics for object-oriented design? Explain class-oriented and object-oriented testing metrics. **(8)**
b) Explain the framework for software technical metrics. How do metrics for analysis and design help improve the software development process? **(8)**

UNIT-IV

Q7. a) Discuss the applications of ERP in functional areas such as marketing, personnel, and production. **(8)**
b) Discuss the implementation life cycle of ERP and the role of training (team training and end-user training) in its success. **(8)**

Q8. a) Explain the components of a Decision Support System (DSS). Discuss its relevance to decision-making in organizations. **(8)**

b) Explain the utility of "What-if" analysis and "Goal- Seek" in decision-making with the help of a practical example. **(8)**

Q9. **Compulsory question:**

a) What is a feasibility study in project management?
b) What is the significance of Gantt charts in project scheduling?
c) What is the purpose of a work breakdown structure (WBS)?
d) Differentiate between size-oriented and function-oriented metrics.
e) Name two technologies associated with ERP.
f) Define sensitivity analysis in the context of DSS.
g) List two characteristics of process groups in project management.
h) Define project milestones. **(8 x 2 =16)**

M. Sc (Honours) in Computer Science

Second Year (Third Semester)

MCS-1911: Soft Computing Techniques using Neural Networks

Time Allowed: 3 Hours **Maximum Marks: 80**

Note: Attempt **five** questions in all, including **Question No. 9** (Unit-V) which is compulsory, and selecting one question from each Unit I-IV.

Unit-I

1.

(a) Define soft computing. Explain how it differs from hard computing with examples. (8 Marks)

(b) Describe the architecture and functionality of artificial neural networks. (8 Marks)

2.

(a) Explain the applications of soft computing in real-life scenarios. (6 Marks)

(b) Discuss various activation functions used in neural networks and their mathematical representation. (10 Marks)

Unit-II

3.

(a) Describe the supervised learning process and its algorithms. (8 Marks)

(b) Write a detailed explanation of the architecture of Hopfield networks. (8 Marks)

4.

(a) Explain the backpropagation algorithm with its mathematical formulation. (10 Marks)

(b) Differentiate between supervised and unsupervised learning with suitable examples. (6 Marks)

Unit-III

5.

(a) Define fuzzy sets and explain their differences from classical sets. (6 Marks)

(b) Discuss various operations performed on fuzzy sets with examples. (10 Marks)

6.

(a) Explain the architecture and application of fuzzy logic control systems. (8 Marks)

(b) Describe the concept of defuzzification and its different methods. (8 Marks)

Unit-IV

7.

(a) Define genetic algorithms and their biological background. (8 Marks)

(b) Explain crossover and mutation operations with examples in genetic algorithms. (8 Marks)

8.

(a) Compare genetic algorithms with traditional optimization methods. (8 Marks)

(b) Discuss the stages of a genetic algorithm with a flowchart. (8 Marks)

Unit-V (Compulsory)

9. Attempt any **four** of the following: $(4 \times 4 = 16$ Marks)

1. Define Hebbian learning and its significance in neural networks.
2. Explain the applications of fuzzy logic systems in control problems.
3. Discuss the difference between genotype and phenotype in genetic algorithms.
4. Highlight the role of selection and fitness functions in genetic algorithms.